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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Consent: Materials and Methods Demographics and Samples Collection Peripheral blood samples from 6 severe burn patients and 6 healthy volunteers were obtained with the written informed consent of participants.
IRB: The study was conducted according to the guidelines of the Declaration of Helsinki, and all procedures were approved by the Human Subjects Review Board of the First Affiliated Hospital of Anhui Medical University (number: PJ2019-06-07).
Field Sample Permit: Burn skin specimens were sampled from a partial-thickness burned area since a full-thickness burned skin undergoing necrosis obviated the detection of gene expressions.
Inclusion and Exclusion Criteria
13 The inclusion criteria of severe burn patients included: thermal injury; % TBSA burn ≥ 60; ABSI score ≥ 10; 18 ≤ age ≤ 80; 4 ≤ days postburn ≤ 7.
Attrition
18 The burn patients were clustered into 3 groups according to the post-burn days (PBD), including PBD 0–3 group (15 patients), PBD 4–7 group (15 patients) and PBD 7–17 group (15 patients), to reveal the temporal gene expression profiles in thermally injured skin.
Sex as a biological variable
Table 1 Demographics for Participants Sampled for NGS and Patients from GSE8056 Research and Sample Type Control Group Sex Age (Years) Mean Age Burn Group Sex Age (Years) Mean Age % TBSA Burn (% Deep Burn) Mean%TBSA ABSI Score Day Postburn NGS of plasma exosome-derived small RNA Normal 01 F 24 28.3 ±5.28 Burn 01 F 54 39 ±16.69 60 (32) 79.83 ±19.99 11 5 Normal 02 F 21 Burn 02 M 28 60 (46) 10 5 Normal 03 F 29 Burn 03 F 25 65 (20) 11 5 Normal 04 M 28 Burn 04 M 54 97 (80) 15 4 Normal 05 M 33 Burn 05 F 54 98 (92) 16 4 Normal 06 M 35 Burn 06 M 19 99 (95) 13 4 cDNA-microarray analysis of skin GSM198875-01 F 35 38.4 ±14.37 GSM198869-01 M 8 30.8 ±14.83 6 14.67 ±12.94 No record 5 GSM198875-02 F 26 GSM198869-02 M 19 8 4 GSM198875-03 F 43 GSM198869-03 M 30 10 4 GSM198875-04 F 20 GSM198869-04 F 11 5 5 GSM198875-05 F 20 GSM198869-05 F 31 15 5 GSM198876-01 F 32 GSM198870-01 M 26 30 4 GSM198876-02 F 43 GSM198870-02 M 33 5 5 GSM198876-03 F 52 GSM198870-03 F 37 4 6 GSM198876-04 F 56 GSM198870-04 F 38 20 7 GSM198876-05 F 44 GSM198870-05 M 18 20 7 GSM198877-01 F 55 GSM198871-01 M 51 50 4 GSM198877-02 M 43 GSM198871-02 M 25 10 5 GSM198877-03 F 22 GSM198871-03 M 42 27 5 GSM198877-04 M 22 GSM198871-04 M 28 5 7 GSM198877-05 F 63 GSM198871-05 M 65 5 6 Abbreviations: TBSA, total body surface area; ABSI, abbreviate burn severity index; NGS, next-generation sequencing; F, female; M, male.

Subject Demographics	
Age: not detected.	
Weight: not detected.	
Randomization	
not detected.	
Blinding	
not detected.	
Power Analysis	
not detected.	
Replication	
not required.	
Data Information	
Identifiers: Table 1 Demographics for Participants Sampled for NGS and Patients from GSE8056 Research and Sample Type Control Group Sex Age (Years) Mean Age Burn Group Sex Age (Years) Mean Age % TBSA Burn (% Deep Burn) Mean%TBSA ABSI Score Day Postburn NGS of plasma exosome-derived small RNA Normal 01 F 24 28.3 ±5.28 Burn 01 F 54 39 ±16.69 60 (32) 79.83 ±19.99 11 5 Normal 02 F 21 Burn 02 M 28 60 (46) 10 5 Normal 03 F 29 Burn 03 F 25 65 (20) 11 5 Normal 04 M 28 Burn 04 M 54 97 (80) 15 4 Normal 05 M 33 Burn 05 F 54 98 (92) 16 4 Normal 06 M 35 Burn 06 M 19 99 (95) 13 4 cDNA-microarray analysis of skin GSM198875-01 F 35 38.4 ±14.37 GSM198869-01 M 8 30.8 ±14.83 6 14.67 ±12.94 No record 5 GSM198875-02 F 26 GSM198869-02 M 19 8 4 GSM198875-03 F 43 GSM198869-03 M 30 10 4 GSM198875-04 F 20 GSM198869-04 F 11 5 5 GSM198875-05 F 20 GSM198869-05 F 31 15 5 GSM198876-01 F 32 GSM198870-01 M 26 30 4 GSM198876-02 F 43 GSM198870-02 M 33 5 5 GSM198876-03 F 52 GSM198870-03 F 37 4 6 GSM198876-04 F 56 GSM198870-04 F 38 20 7 GSM198876-05 F 44 GSM198870-05 M 18 20 7 GSM198877-01 F 55 GSM198871-01 M 51 50 4 GSM198877-02 M 43 GSM198871-02 M 25 10 5 GSM198877-03 F 22 GSM198871-03 M 42 27 5 GSM198877-04 M 22 GSM198871-04 M 28 5 7 GSM198877-05 F 63 GSM198871-05 M 65 5 6 Abbreviations: TBSA, total body surface area; ABSI, abbreviate burn severity index; NGS, next-generation sequencing; F, female; M, male.	GSE8056

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Primary antibodies employed included rabbit anti-human CD63 antibody (EXOAB-CD63A-1, SBI, USA), and rabbit anti-human Calnexin antibody (YT0613, ImmunoWay, USA).	anti-human CD63		
	anti-human Calnexin antibody (YT0613		
Goat anti-rabbit HRP antibody (SBI, USA) was used as a secondary antibody.	anti-rabbit HRP		
Software and Algorithms			
Table 1 Demographics for Participants Sampled for NGS and Patients from GSE8056 Research and Sample Type Control Group Sex Age (Years) Mean Age Burn Group Sex Age (Years) Mean Age % TBSA Burn (% Deep Burn) Mean%TBSA ABSI Score Day Postburn NGS of plasma exosome-derived small RNA Normal 01 F 24 28.3 ±5.28 Burn 01 F 54 39 ±16.69 60 (32) 79.83 ±19.99 11 5 Normal 02 F 21 Burn 02 M 28 60 (46) 10 5 Normal 03 F 29 Burn 03 F 25 65 (20) 11 5 Normal 04 M 28 Burn 04 M 54 97 (80) 15 4 Normal 05 M 33 Burn 05 F 54 98 (92) 16 4 Normal 06 M 35 Burn 06 M 19 99 (95) 13 4 cDNA-microarray analysis of skin GSM198875-01 F 35 38.4 ±14.37 GSM198869-01 M 8 30.8 ±14.83 6 14.67 ±12.94 No record 5 GSM198875-02 F 26 GSM198869-02 M 19 8 4 GSM198875-03 F 43 GSM198869-03 M 30 10 4 GSM198875-04 F 20 GSM198869-04 F 11 5 5 GSM198875-05 F 20 GSM198869-05 F 31 15 5 GSM198876-01 F 32 GSM198870-01 M 26 30 4 GSM198876-02 F 43 GSM198870-02 M 33 5 5 GSM198876-03 F 52 GSM198870-03 F 37 4 6 GSM198876-04 F 56 GSM198870-04 F 38 20 7 GSM198876-05 F 44 GSM198870-05 M 18 20 7 GSM198877-01 F 55 GSM198871-01 M 51 50 4 GSM198877-02 M 43 GSM198871-02 M 25 10 5 GSM198877-03 F 22 GSM198871-03 M 42 27 5 GSM198877-04 M	NGS		Suggestion: NGS-SNP (RRID:SCR_005182)(link)

22 GSM198871-04 M 28 5 7 GSM198877-05 F 63 GSM198871-05 M 65 5 6 Abbreviations: TBSA, total body surface area; ABSI, abbreviate burn severity index; NGS, next-generation sequencing; F, female; M, male.			
Subsequent RNA-seq reads were mapped to miRbase, piRNAbank and Rfam databases.	piRNAbank		Suggestion: piRNABank (RRID:SCR_007858)(link)
	Rfam		
Expression matrix of quantified UMI counts of miRNAs was normalized to counts per million (CPM) and calculated to relative log expression via the EdgeR package.	EdgeR		
Target Genes Prediction and Enrichment Analyses Target genes of DEMs were predicted in mirDIP (http://ophid.utoronto.ca/mirDIP/), which integrated information of miRNAs and genes in Homo sapiens from 30 major databases including miRbase, miRDB and TargetScan.	mirDIP		
	http://ophid.utoronto.ca/mirDIP/		
	miRbase		
	TargetScan		Suggestion: TargetScan (RRID:SCR_010845)(link)
Enrichment analyses of GO and KEGG were conducted using the ClusterProfiler17 package in R software (version 4.1.2).	ClusterProfiler17		
Differentially expressed genes (DEGs) were identified using the LIMMA19 package with enrichment analyses conducted using the ClusterProfiler package in R software.	ClusterProfiler		
Enrichment analyses of GO and KEGG were repeated on the focus genes.	KEGG		
Construction of PPI and miRNA-mRNA Networks A protein–protein interaction (PPI) network of the focus genes was constructed in Cytoscape software (version 3.7.2) based on the STRING database (https://www.string-db.org/).20 Hub genes in the PPI network were selected by degree using the cytoHubba21 plugin of Cytoscape.	Cytoscape		
	STRING		Suggestion: STRING (RRID:SCR_005223)(link)

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